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***The Digital Blackout: An Econometric Analysis of Internet
Connectivity Shocks and Exchange Rate Volatility in Iran***

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Warsaw, Poland

2026

Abstract

This study examines the impact of internet connectivity shocks on exchange-rate dynamics in Iran during the nationwide internet shutdown of January 2026. Using daily observations from January to March 2026, the analysis combines network telemetry data from the Internet Outage Detection and Analysis (IODA) platform with free-market USD/Toman¹ exchange-rate data obtained from Bonbast². Treating the shutdown as an exogenous digital shock, the study investigates whether disruptions in internet connectivity contributed to the depreciation of the Iranian currency.

The empirical analysis employs Augmented Dickey-Fuller (ADF) unit root tests, Ordinary Least Squares (OLS) regression, and an Autoregressive Distributed Lag (ARDL) framework to examine both the immediate and persistent effects of connectivity loss on exchange-rate movements. By integrating network measurement data with financial market data, the research provides a novel approach to assessing the macroeconomic consequences of digital isolation.

The findings contribute to the emerging literature on digital economics and financial stability by offering empirical evidence on the relationship between internet disruptions and currency market behavior in a sanctioned and politically sensitive economy. The proposed framework may also serve as a foundation for future research on the economic effects of internet shutdowns and digital infrastructure shocks.

1. Introduction

The increasing integration of digital technologies into economic activity has transformed the role of internet infrastructure from a communication tool into a critical component of financial markets, trade, and information exchange. Modern currency markets depend on the rapid dissemination of information, allowing market participants to continuously update expectations and respond to economic and political developments. Exchange-rate research increasingly recognizes that information flows play a fundamental role in price formation, particularly in

¹ A unit of monetary value equal to 10 Iranian rials, <https://www.merriam-webster.com/dictionary/toman>

² Live exchange rates in Iran's free market, <https://www.merriam-webster.com/dictionary/toman>

environments characterized by uncertainty and limited market transparency. Evans and Lyons (2002) argue that exchange-rate movements are influenced not only by macroeconomic fundamentals but also by the transmission of information through financial markets, suggesting that disruptions to information channels may affect currency valuation³.

Internet shutdowns represent one of the most extreme forms of information disruption. By restricting access to communication networks, online financial services, digital commerce, and real-time market information, such disruptions may generate economic consequences beyond their immediate technological effects. Existing studies have primarily examined the impact of internet connectivity on economic growth, trade performance, and productivity (Mattes, 2012; Shahhosseini, 2021). Similarly, research on internet shutdowns has focused largely on their social, political, and economic costs, including business interruption, reduced digital activity, and losses in economic output. However, relatively little attention has been given to the relationship between internet connectivity shocks and exchange-rate dynamics.⁴

Iran provides a particularly relevant case for investigating this relationship. The Iranian economy has experienced persistent inflation, exchange-rate instability, and international sanctions, making market expectations highly sensitive to information availability and economic uncertainty. Previous research has shown that internet access plays a significant role in Iran's economic activity and international trade, while exchange-rate fluctuations remain a key determinant of economic performance (Shahhosseini, 2021)⁵. In such an environment, disruptions to digital communication may amplify uncertainty, increase information asymmetry, and affect the behavior of market participants in the foreign-exchange market.

The nationwide internet shutdown that occurred in Iran during January 2026 offers a unique opportunity to examine these mechanisms. Recent network measurement research documented the event using multiple independent measurement systems

³ Evans, M. D. D., & Lyons, R. K. (2002). Order flow and exchange rate dynamics. *Journal of Political Economy*

⁴ Mattes, A. (2012). *Goods Follow Bytes: The Impact of ICT on EU Trade*. DIW Berlin Discussion
Shahhosseini, M. (2021). *The Impact of the Internet on Iran's Bilateral Trade: Modified Gravity Model Approach*. Iranian Journal of Trade Studies

⁵ Shahhosseini, M. (2021). The Impact of the Internet on Iran's Bilateral Trade: Modified Gravity Model Approach. Iranian Journal of Trade Studies

and reported substantial reductions in internet reachability across the country. The study *A Multi-Perspective Study of the Internet Shutdown in Iran* demonstrated that the disruption affected a large proportion of Iran's internet infrastructure and represented one of the most significant connectivity interruptions observed in recent years⁶. While this literature provides valuable evidence regarding the technical characteristics and scale of the shutdown, its macroeconomic implications remain largely unexplored.

This gap is particularly important because internet shutdowns can be viewed as exogenous shocks to information flows. If exchange-rate markets rely on timely access to information, a sudden and severe reduction in connectivity may influence currency valuation by disrupting digital arbitrage, limiting access to market information, and increasing uncertainty among investors and households.

This study addresses this gap by investigating the relationship between internet connectivity shocks and the free-market USD/Toman exchange rate during the January–March 2026 period.

2. Literature Review

2.1 Internet Connectivity and Economic Performance

The expansion of internet infrastructure has fundamentally changed the way economic activity is organized and coordinated. As digital networks have become increasingly integrated into trade, finance, and business operations, internet connectivity has evolved from a communication technology into a productive economic asset. Firms rely on digital networks to access market information, coordinate supply chains, conduct transactions, and interact with customers across geographical boundaries. Consequently, disruptions to connectivity can affect not only communication but also broader economic outcomes.

A substantial body of literature has examined the relationship between information and communication technologies (ICT) and economic performance. Mattes (2012) argues that improvements in ICT infrastructure facilitate international trade by reducing information and transaction costs between trading partners. Using data

⁶ Jahromi, A. S., & Jaskolka, J. (2026). A multi-perspective study of the Internet shutdown in Iran

from European Union member states, the study finds that digital connectivity strengthens trade relationships and contributes to economic integration, particularly in sectors where information exchange is critical. These findings support the view that internet infrastructure generates economic value by lowering barriers to market participation and improving access to information⁷.

Similar conclusions have been reached in studies focusing on developing and emerging economies. Shahhosseini (2021), examining Iran's bilateral trade relationships through a modified gravity model, finds that internet penetration has a statistically significant positive effect on trade flows. The study suggests that internet access enhances economic interaction by improving communication efficiency and reducing informational friction between trading partners. Importantly, the analysis also identifies exchange-rate conditions as a significant determinant of trade performance, highlighting the close relationship between digital connectivity and broader macroeconomic outcomes⁸.

Research conducted by the Organisation for Economic Co-operation and Development (OECD) further emphasizes the economic importance of digital infrastructure. OECD reports describe connectivity as a key enabler of productivity growth, market efficiency, and digital transformation. As economic activity becomes increasingly dependent on real-time information and online transactions, the costs associated with connectivity disruptions become more significant. Interruptions to digital infrastructure can reduce market efficiency, delay information transmission, and increase uncertainty among economic actors⁹.

Collectively, these studies suggest that internet connectivity influences economic performance through its effect on information flows. By reducing information asymmetries and facilitating faster dissemination of market-relevant information, digital networks contribute to more efficient economic decision-making. This insight is particularly relevant for financial markets, where prices react continuously to new information. If internet access facilitates the transmission of

⁷ Mattes, A. (2012). *Goods Follow Bytes: The Impact of ICT on EU Trade*. DIW Berlin Discussion

⁸ Shahhosseini, M. (2021). *The Impact of Internet on Iran's Bilateral Trade: Modified Gravity Model Approach*. Iranian Journal of Trade Studies

⁹ OECD. (2019). *Measuring the Digital Transformation: A Roadmap for the Future*. OECD Publishing.

information, then large-scale disruptions to connectivity may alter market behavior and contribute to economic instability.

2.2 Internet Shutdowns as Economic Shocks

Internet shutdowns are deliberate disruptions of digital communication networks that restrict access to online services, information, and communication channels. While governments often justify such measures on security or political grounds, researchers increasingly recognize them as economic shocks capable of affecting business activity, market efficiency, and information exchange. The growing frequency of internet shutdowns worldwide has attracted attention from both policymakers and academics, particularly as economic activity becomes increasingly dependent on digital infrastructure.

Existing research suggests that internet disruptions generate substantial economic costs. Studies by the Internet Society indicate that shutdowns reduce economic activity by interrupting e-commerce, business operations, financial transactions, and access to digital services. Similarly, OECD reports emphasize that digital connectivity has become a critical component of modern economic systems, making disruptions increasingly costly as economies undergo digital transformation.¹⁰

Iran provides an important case study in this literature. In January 2026, the country experienced one of the most extensive nationwide internet shutdowns documented to date. Using active probing, passive network measurements, and Border Gateway Protocol (BGP) analysis, Jahromi and Jaskolka (2026) found that between 96.5% and 97.4% of observable Iranian network prefixes became unreachable during the disruption, indicating a highly centralized and coordinated shutdown mechanism. Their findings demonstrate that the event represented a large-scale disruption to the country's digital infrastructure and information environment.¹¹

Although recent studies have significantly improved understanding of how internet shutdowns are implemented and measured, their focus remains largely on technical

¹⁰ OECD. (2019). *Measuring the Digital Transformation: A Roadmap for the Future*. OECD Publishing.

¹¹ Jahromi, A. S., & Jaskolka, J. (2026). *A multi-perspective study of the Internet shutdown in Iran*

characteristics, censorship mechanisms, and direct economic costs. Consequently, relatively little is known about how large-scale connectivity disruptions influence financial markets and exchange-rate behavior. This gap is particularly relevant in economies such as Iran. The January 2026 shutdown therefore provides a unique opportunity to examine whether disruptions to digital connectivity can function as measurable macroeconomic shocks.

2.3 Information Asymmetry and Exchange Rate Dynamics

Traditional exchange-rate models explain currency movements through macroeconomic fundamentals such as inflation, interest rates, money supply, and trade balances. While these factors remain important, they are often insufficient to explain short-run exchange-rate fluctuations. This limitation led to the development of exchange-rate microstructure theory, which emphasizes the role of information transmission in financial markets.

The importance of information in financial markets is also emphasized by Mankiw and Reis (2002) through their Sticky Information model. Unlike traditional models that assume economic agents continuously update their expectations, Mankiw and Reis argue that information is incorporated gradually because individuals and firms do not instantly acquire or process new information. Consequently, delays in information transmission can generate temporary market inefficiencies and increase uncertainty. Although developed in a broader macroeconomic context, this perspective suggests that disruptions to information flows may influence the behavior of financial markets.¹²

Internet connectivity plays an increasingly important role in this process. Digital networks facilitate access to financial information, market prices, economic news, and communication between market participants. In modern foreign-exchange markets, information is transmitted almost instantaneously through digital channels, allowing traders and investors to respond rapidly to new developments. Consequently, large-scale internet disruptions may interfere with information dissemination and reduce market transparency.

¹² Mankiw, N. G., & Reis, R. (2002). Sticky information versus sticky prices: A proposal to replace the New Keynesian Phillips Curve. *Quarterly Journal of Economics*

From a theoretical perspective, internet shutdowns can increase information asymmetry by limiting access to market information and restricting communication between economic agents. Information asymmetry arises when some market participants possess information that is unavailable to others, leading to less efficient price formation and increased uncertainty (Akerlof, 1970).¹³ During a widespread internet disruption, access to exchange-rate quotations, financial news, online trading platforms, and digital communication channels may become restricted. As a result, market participants may face greater difficulty in assessing economic conditions and forming expectations regarding future currency movements.

These mechanisms may be particularly relevant in Iran. Previous research has shown that exchange-rate movements in Iran are strongly influenced by expectations, political uncertainty, sanctions, and inflationary pressures. In such an environment, informal information channels often play an important role in shaping market sentiment. A sudden reduction in internet connectivity may therefore amplify uncertainty and influence exchange-rate behavior by disrupting the flow of information that market participants use to form expectations.

Building on the theories of information asymmetry and exchange-rate microstructure, this study hypothesizes that large-scale internet disruptions may affect the valuation of the Iranian Toman by reducing information availability, increasing uncertainty, and weakening market efficiency. The January 2026 internet shutdown therefore provides a unique opportunity to examine whether disruptions to digital infrastructure are associated with measurable changes in exchange-rate dynamics.

2.4 Econometric Literature for Time-Series Analysis

The empirical part of this study relies on standard time-series econometric methods because both internet connectivity loss and exchange-rate movements are observed over time. In time-series analysis, testing for stationarity is an essential first step, since regressions using non-stationary variables may produce misleading relationships. Dickey and Fuller (1979) developed the unit root testing framework

¹³ Akerlof, G. A. (1970). The market for "lemons": Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*

that is commonly used to determine whether a series is stationary or requires transformation. In this study, the Augmented Dickey-Fuller test is applied before estimating the relationship between internet loss and exchange-rate changes¹⁴.

After transforming the variables into stationary form, Ordinary Least Squares regression is used as a baseline model to estimate the immediate association between changes in connectivity loss and changes in the USD/Toman exchange rate. Although OLS provides a simple starting point, time-series data often contain lagged effects. For this reason, the study also uses Granger causality testing, originally developed by Granger (1969), to examine whether past values of internet connectivity loss help predict future exchange-rate movements.¹⁵

To capture both immediate and delayed effects, the analysis further applies an Autoregressive Distributed Lag framework. The ARDL approach developed by Pesaran, Shin, and Smith (2001) is widely used in applied time-series research because it allows the dependent variable to respond to both its own past values and lagged values of explanatory variables. This is especially relevant for the present study, where the effect of an internet shutdown may not be limited to a single day but may persist through market expectations and delayed adjustment.¹⁶

The literature reviewed above highlights three key insights. First, internet connectivity plays an increasingly important role in economic activity and information transmission. Second, internet shutdowns represent measurable disruptions to digital infrastructure with potentially significant economic consequences. Third, exchange-rate literature emphasizes the importance of information flows and market expectations in currency valuation. The following chapter presents the data sources, variable construction, and econometric methodology used to examine the impact of the January 2026 internet shutdown on the USD/Toman exchange rate in Iran.

¹⁴ Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*,

¹⁵ Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods.

¹⁶ Pesaran, M. H., Shin, Y., & Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationships. *Journal of Applied Econometrics*

3. Data and Methodology

3.1 Data Sources

This study investigates the relationship between internet connectivity shocks and exchange-rate movements in Iran during the nationwide internet shutdown of January 2026. To achieve this objective, two datasets were combined: internet connectivity data obtained from the Internet Outage Detection and Analysis (IODA) platform¹⁷ and free-market USD/Toman exchange-rate data obtained from Bonbast¹⁸.

IODA is a network measurement platform developed by the Georgia Institute of Technology that provides real-time indicators of internet reachability and network availability. The platform aggregates information from multiple measurement systems and is widely used to identify large-scale internet outages and disruptions. In this study, the Active Probing metric was selected because it provides a direct measure of observable internet reachability across Iranian network infrastructure.

The exchange-rate dataset was obtained from Bonbast, a widely used source for free-market foreign-exchange rates in Iran. Daily USD/Toman observations were collected for the same period as the network data. The analysis focuses on the free-market exchange rate because it more accurately reflects market expectations and real-time currency valuation than official exchange rates.

The sample covers the period from 1 January 2026 to 31 March 2026, resulting in approximately 90 daily observations after data processing and synchronization.

3.2 Data Preparation and Variable Construction

The IODA dataset was originally reported at approximately 30-minute intervals. To ensure consistency with the daily exchange-rate observations, the connectivity data were aggregated into daily averages. The daily Active Probing measure was then transformed into a connectivity-loss indicator using the following relationship:

¹⁷ Internet Outage Detection and Analysis (IODA) platform, <https://ioda.inetintel.cc.gatech.edu/>

¹⁸ Live exchange rates in Iran's free market, <https://www.merriam-webster.com/dictionary/toman>

$$\text{Loss}_t = 100 - \text{Active Probe}$$

where Loss_t represents the percentage loss of internet connectivity on day (t).

The exchange-rate series was transformed into logarithmic returns to reduce non-stationarity and allow interpretation in terms of relative changes. The transformed variable is defined as:

$$\Delta \log(\text{ExRate}_t) = \ln(\text{ExRate}_t) - \ln(\text{ExRate}_{t-1})$$

Similarly, the connectivity-loss variable was transformed into first differences:

$$d_Loss_t = \text{Loss}_t - \text{Loss}_{t-1}$$

These transformations produce stationary series suitable for time-series econometric analysis.

3.3 Descriptive Analysis

Prior to estimation, the data were examined using summary statistics and graphical analysis. Time-series plots were used to visualize the evolution of the USD/Toman exchange rate and internet connectivity loss during the study period. Particular attention was given to identifying abrupt changes associated with the January 2026 shutdown event.

Correlation analysis was also conducted to provide an initial assessment of the relationship between exchange-rate movements and connectivity loss. A correlation matrix and heatmap were generated to illustrate the strength and direction of the association between the variables.

3.4 Econometric Framework

The empirical analysis was conducted using a sequence of time-series econometric techniques designed to evaluate both contemporaneous and dynamic relationships between internet connectivity loss and exchange-rate movements.

3.4.1 Stationarity Testing

The Augmented Dickey-Fuller (ADF) test was applied to both the original and transformed series to determine whether the variables contained unit roots. Testing for stationarity is necessary because regressions involving non-stationary variables may produce spurious results.

3.4.2 Ordinary Least Squares Regression

A baseline Ordinary Least Squares (OLS) regression was estimated to examine the immediate relationship between changes in internet connectivity loss and changes in the USD/Toman exchange rate. The model is specified as:

$$d_log_ExRate_t = \alpha + \beta d_Loss_t + \epsilon_t$$

where $(d_log_ExRate_t)$ denotes daily exchange-rate returns and (d_Loss_t) represents daily changes in internet connectivity loss.

3.4.3 Granger Causality Analysis

To examine predictive relationships between the variables, a Granger causality test was performed. The test evaluates whether past changes in internet connectivity loss contain information that improves the prediction of future exchange-rate movements.

3.4.4 Autoregressive Distributed Lag Model

To investigate whether the effects of internet disruptions persist over time, an Autoregressive Distributed Lag (ARDL) model was estimated. The model allows exchange-rate returns to depend on both their own lagged values and lagged changes in connectivity loss:

$$d_log_ExRate_t = \alpha + \phi d_log_ExRate_{t-1} + \beta_0 d_Loss_t + \beta_1 d_Loss_{t-1} + \varepsilon_t$$

This specification captures both immediate and delayed effects of internet connectivity shocks on currency-market behavior.

3.4.5 Structural Break Analysis

Given the sudden nature of the January 2026 internet shutdown, structural break analysis was performed to identify potential changes in the relationship between connectivity loss and exchange-rate movements during the sample period.

3.5 Diagnostic Tests

Several diagnostic tests were conducted to evaluate the adequacy of the estimated models. The Breusch-Godfrey test was used to detect serial correlation, the Breusch-Pagan test was employed to assess heteroskedasticity, and the Jarque-Bera test was used to evaluate the normality of residuals. These procedures help ensure the reliability and robustness of the econometric results presented in the following chapter.

4. Results and Discussion

This chapter presents the empirical findings of the study. The analysis begins with descriptive statistics and graphical examination of the exchange-rate and internet connectivity series. Stationarity tests are then conducted to determine the appropriate transformations for time-series analysis. The chapter subsequently reports the results of the Ordinary Least Squares (OLS) regression, diagnostic tests, Granger causality analysis, and the Autoregressive Distributed Lag (ARDL) model. Together, these methods provide evidence regarding both the immediate and delayed effects of internet connectivity shocks on exchange-rate dynamics in Iran.

4.1 Descriptive Statistics

Table 4.1 presents summary statistics for the variables included in the analysis. The sample consists of daily observations covering the period from 1 January to 31 March 2026.

Table 1: Statistics Summary of the Data

Statistic	Date	ExRate	ActiveProbe	Loss
Min.	2026-01-01	135,050	2.893	1.158
1st Qu.	2026-01-23	146,375	3.184	4.578
Median	2026-02-14	153,950	3.778	96.222
Mean	2026-02-14	153,209	44.120	55.880
3rd Qu.	2026-03-08	160,175	95.422	96.816
Max.	2026-03-31	172,050	98.842	97.107

Note: This table provides the summary statistics for the exchange rate (ExRate), active probing connectivity metrics (ActiveProbe), and the calculated percentage of internet connectivity loss (Loss) for the period of Q1 2026.

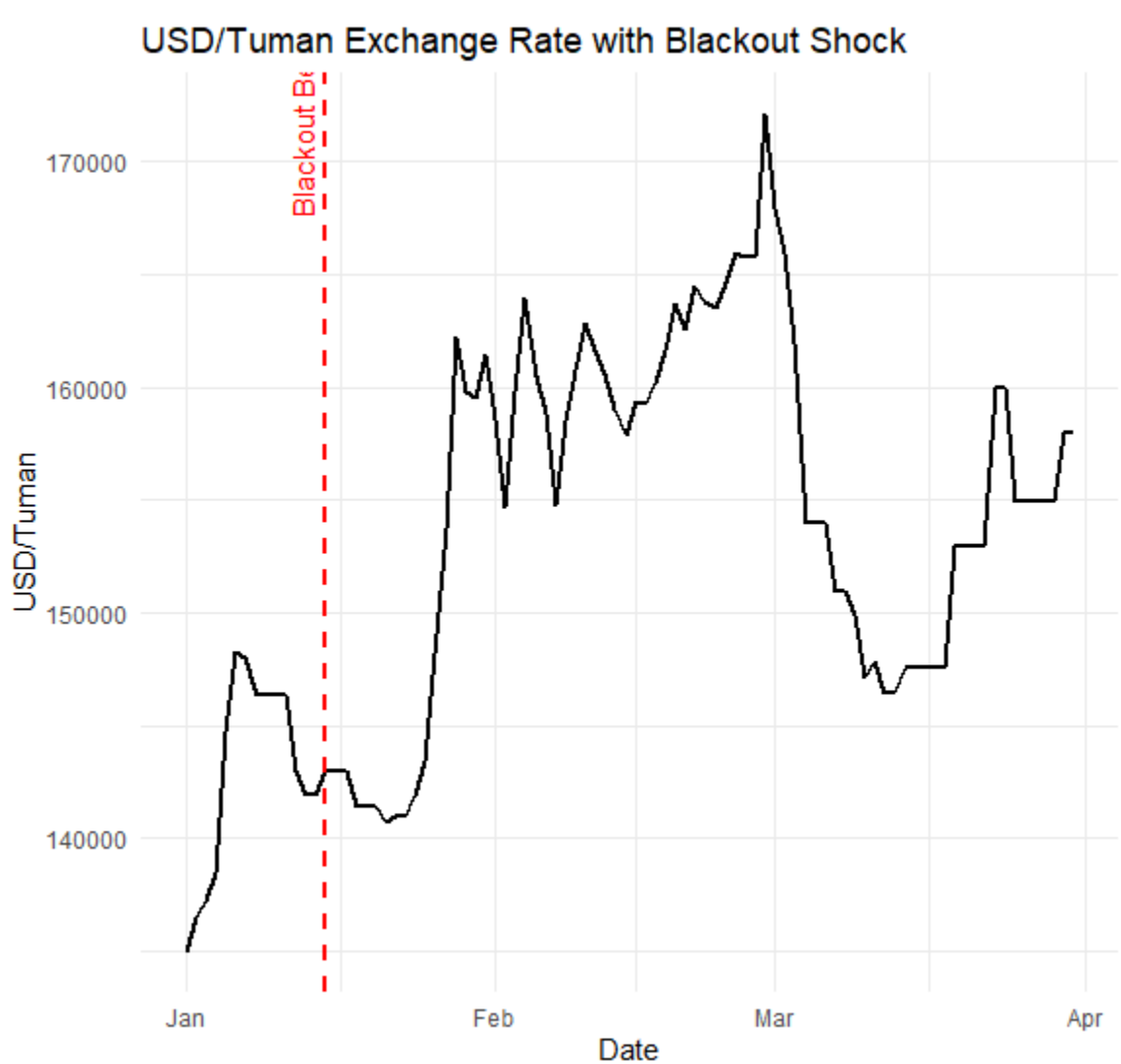
The exchange rate ranged from 135,050 to 172,050 IRR per USD, indicating the substantial variation in both exchange-rate movements and connectivity loss suggests that the dataset captures periods of both normal internet activity and severe network disruption. Consequently, the sample provides a suitable basis for examining whether internet connectivity shocks are associated with exchange-rate dynamics.

The correlation coefficient between exchange-rate returns and changes in internet loss is -0.12, indicating only a weak contemporaneous relationship. This preliminary finding suggests that the impact of internet disruptions may not occur immediately and motivates the use of dynamic time-series methods.

4.2 Graphical Analysis

Figure 4.1 illustrates the evolution of the free-market USD/Toman exchange rate during the first quarter of 2026. Following the onset of the nationwide internet disruption, the exchange rate exhibits greater volatility and a pronounced upward trend. The Iranian currency depreciated from approximately 135,000 IRR per USD in early January to levels exceeding 170,000 IRR per USD by early March.

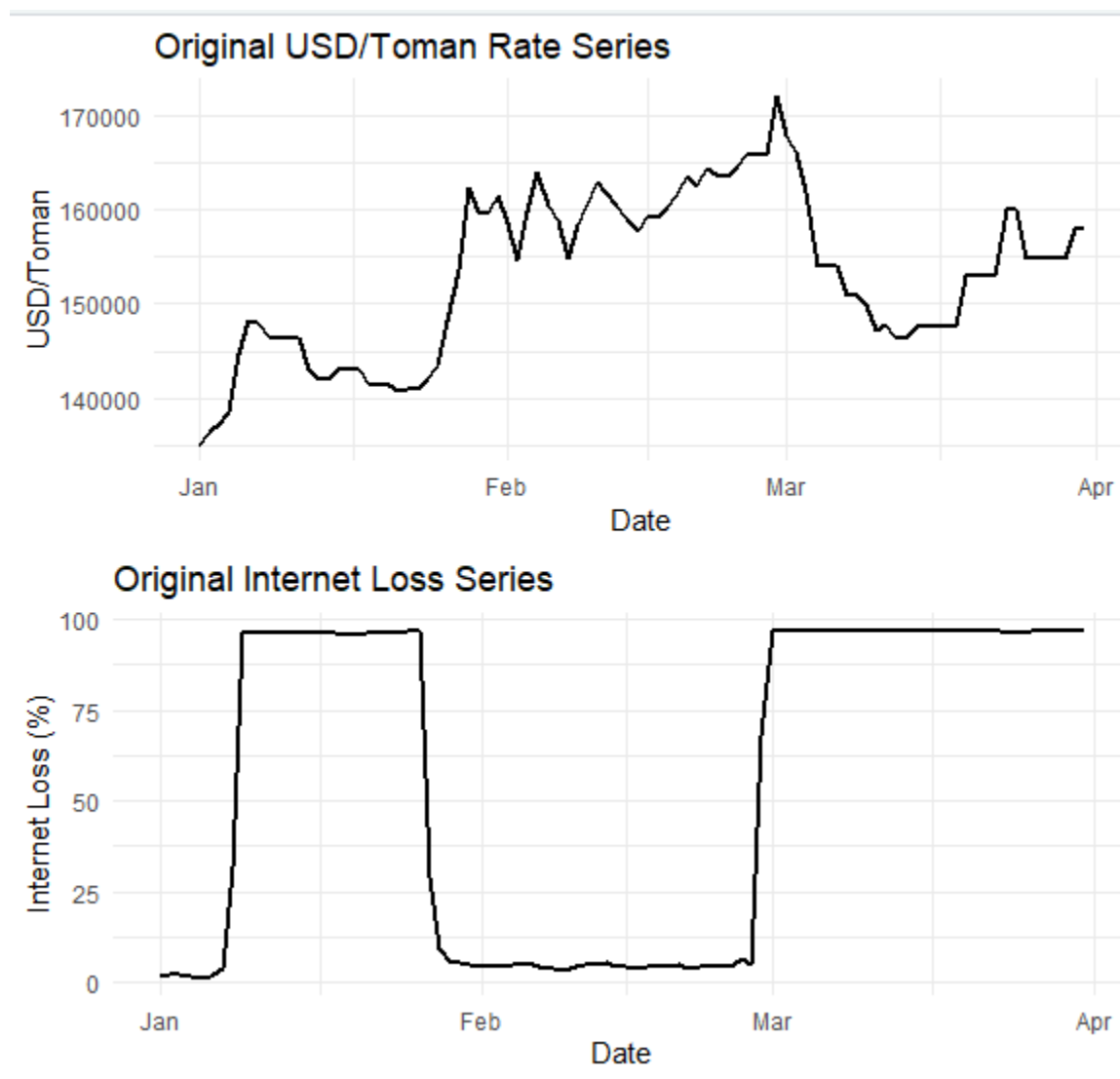
Figure 4.1: USD/Toman Exchange Rate - January 2026 Internet Shutdown



Note: The vertical dashed line indicates the beginning of the January 2026 nationwide internet shutdown. Exchange-rate data represent daily free-market USD/Toman observations obtained from Bonbast.

Figure 4.2 presents the corresponding internet-loss series derived from IODA network measurements. The data reveal two major disruption episodes characterized by near-total connectivity loss. Between these periods, connectivity remained relatively stable. The abrupt nature of these changes suggests that internet disruption occurred as a discrete shock rather than as a gradual deterioration of network performance.

Figure 4.2: USD/Toman Exchange Rate and Internet Loss Series



Note: The upper panel displays the daily free-market USD/Toman exchange rate obtained from Bonbast. The lower panel presents the internet-loss indicator derived from IODA Active Probing data, calculated as $100 - \text{Active Probe100}$. Source: Author's calculations based on Bonbast and IODA data.

A visual comparison of the two series indicates that periods of severe internet disruption coincide with heightened exchange-rate volatility and stronger depreciation pressure on the Iranian currency. Although graphical evidence alone cannot establish causality, the observed timing provides preliminary support for the hypothesis that internet shutdowns contributed to financial-market instability.

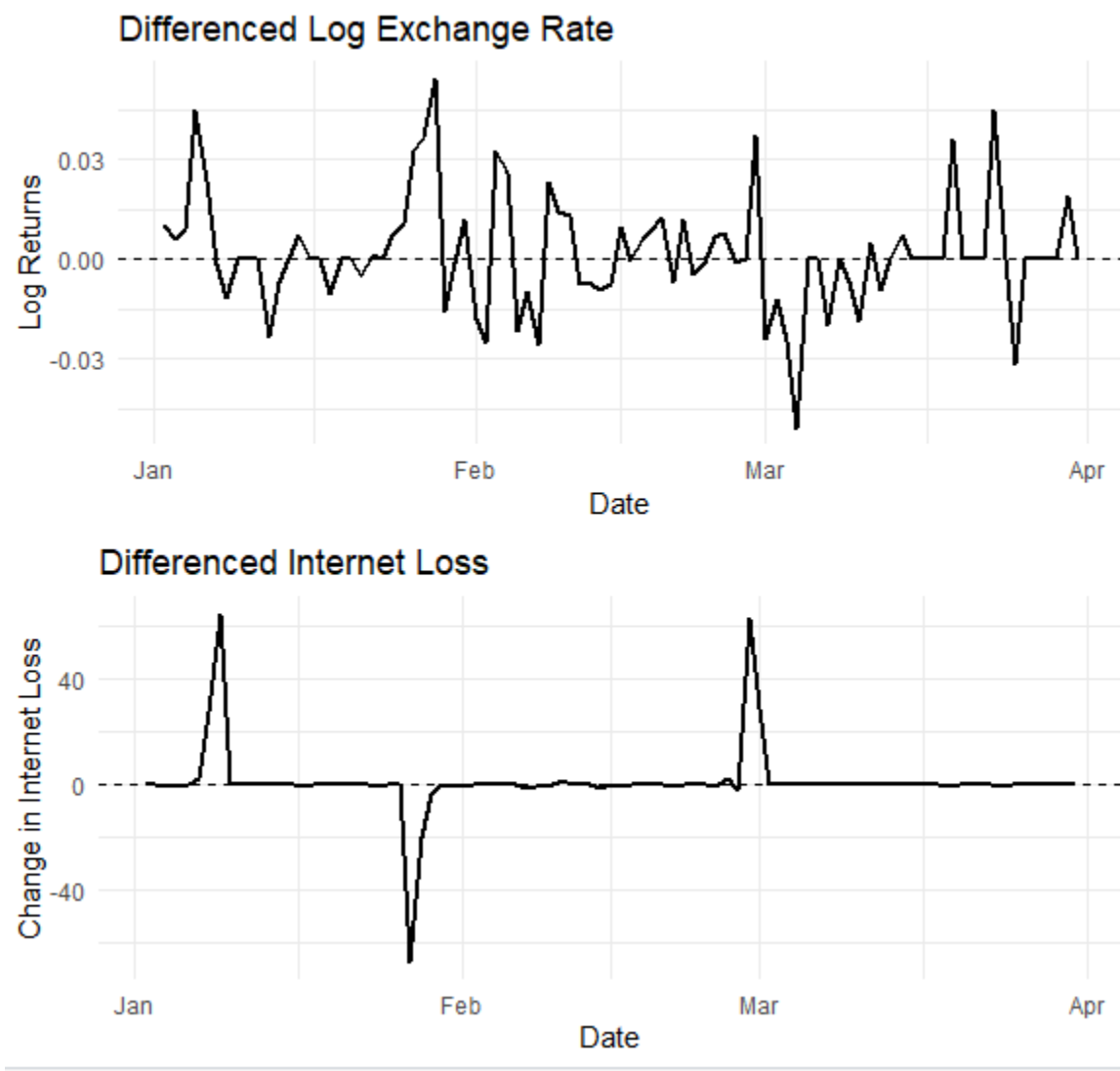
4.3 Stationarity Analysis

Prior to estimation, stationarity tests were performed using the Augmented Dickey-Fuller (ADF) procedure.

The internet-loss series produced an ADF test statistic of -2.020 with a p-value of 0.568. Consequently, the null hypothesis of a unit root could not be rejected, indicating that the series was non-stationary in levels. Similar behavior was observed for the exchange-rate series.

To address this issue, the exchange rate was transformed into logarithmic returns and the internet-loss variable was first-differenced. Figure 4.3 presents the transformed series. Both variables fluctuate around a stable mean and no longer exhibit the persistent trends observed in the original data. These transformations produced stationary series suitable for time-series econometric analysis.

Figure 4.3: Transformed Stationary Series



Note: The upper panel displays the differenced logarithmic exchange rate. The lower panel displays the first-differenced internet loss series (d_Loss_t). The variables are now stationary, making them suitable for time-series econometric analysis. Source: Author's calculations based on Bonbast and IODA data.

4.4 Ordinary Least Squares Regression Results

Table 4.2 reports the results of the baseline OLS regression examining the relationship between changes in internet connectivity loss and exchange-rate returns.

Table 4.2: Ordinary Least Squares Regression Results

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.00193	0.00182	1.059	0.293
d_Loss	-0.00016	0.00014	-1.131	0.261

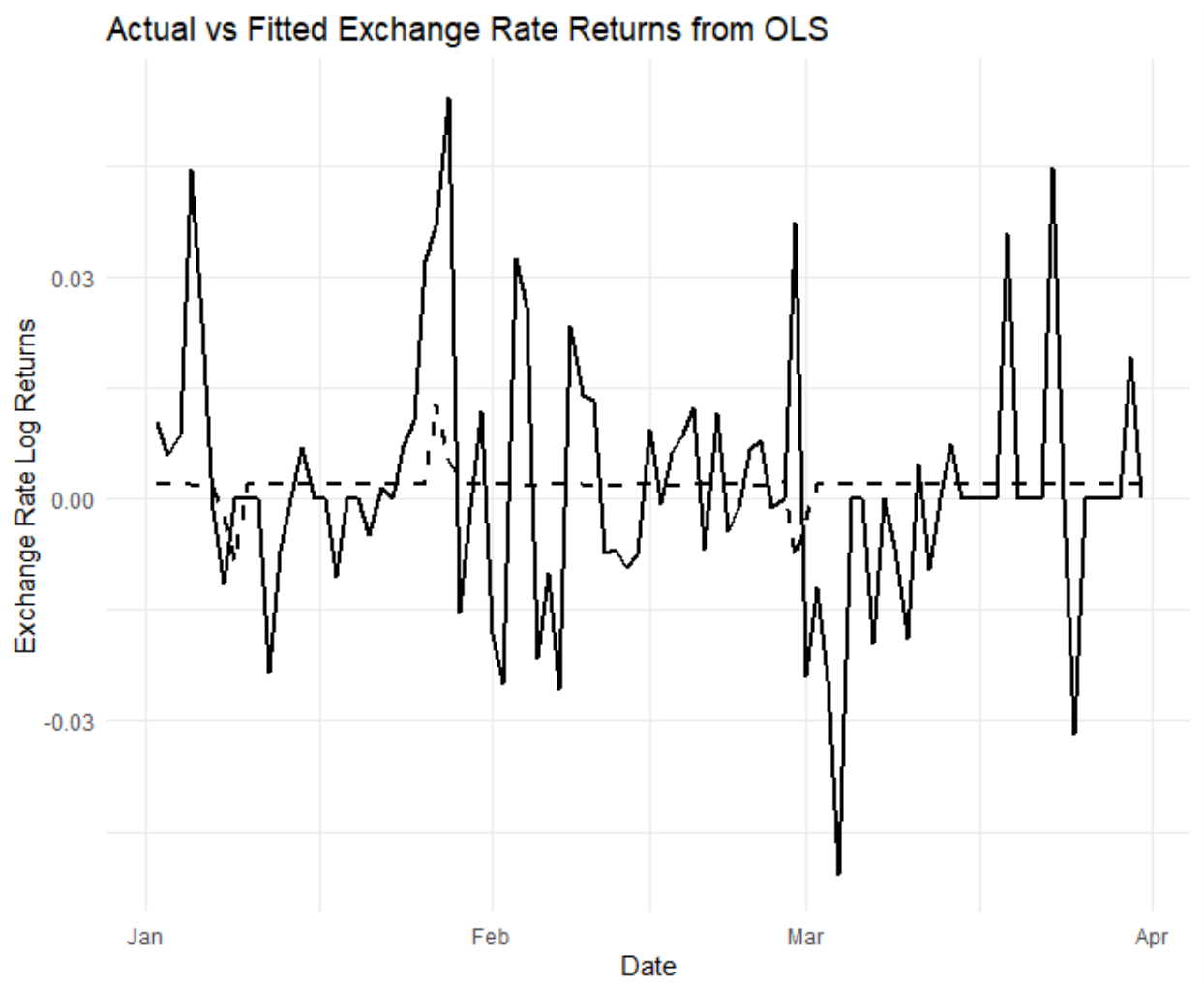
Note: Multiple R-squared: 0.01448, Adjusted R-squared: 0.00315, F-statistic: 1.278 (p-value: 0.2614).

The coefficient associated with internet-loss changes is negative ($= -0.00016$), suggesting that increases in connectivity loss are associated with lower exchange-rate returns. However, the coefficient is not statistically significant ($p = 0.261$). Therefore, the null hypothesis that internet-loss changes have no immediate effect on exchange-rate returns cannot be rejected.

The model explains only a small proportion of exchange-rate variation ($R^2 = 0.014$). This indicates that internet connectivity loss alone is insufficient to explain short-run exchange-rate fluctuations.

Figure 4.4 compares actual exchange-rate returns with the fitted values generated by the OLS model. The fitted series remains close to zero throughout the sample period and fails to replicate the magnitude and timing of observed exchange-rate movements. This result reinforces the conclusion that the relationship between internet connectivity loss and exchange-rate returns is not adequately captured by a simple contemporaneous linear model.

Figure 4.4: Actual vs. Fitted Exchange-Rate Returns (OLS Model)



Note: This figure compares the actual daily exchange-rate returns with the fitted values generated by the OLS model. The fitted series remains close to zero throughout the sample period and fails to replicate the magnitude and timing of the observed exchange-rate movements, consistent with the low R^2 value of the model.

4.5 Diagnostic Tests

Several diagnostic tests were conducted to assess the adequacy of the OLS specification.

The Breusch-Godfrey test produced a p-value of 0.249, indicating no statistically significant autocorrelation in the residuals. Similarly, the Breusch-Pagan test

generated a p-value of 0.614, providing no evidence of heteroskedasticity. These results suggest that the residuals satisfy two important assumptions of the OLS framework.

The Jarque-Bera test yielded a p-value of 0.003, leading to rejection of the null hypothesis of normally distributed residuals. This finding indicates the presence of skewness, excess kurtosis, or extreme observations. Such behavior is common in financial time-series data, particularly exchange-rate returns, which frequently exhibit volatility clustering and non-normal distributions.

4.6 Granger Causality Results

To investigate whether internet connectivity shocks contain predictive information regarding future exchange-rate movements, a Granger causality test was performed.

The test produced an F-statistic of 4.228 with a p-value of 0.018. Since the p-value is below the 5% significance level, the null hypothesis that internet-loss changes do not Granger-cause exchange-rate returns is rejected.

This finding suggests that past changes in internet connectivity contain statistically significant information about future exchange-rate movements. Importantly, this result differs from the OLS findings, which identified no significant contemporaneous relationship. The evidence therefore indicates that the effects of internet disruption may emerge gradually rather than instantaneously.

4.7 Autoregressive Distributed Lag (ARDL) Results

Table 4.3 reports the results of the ARDL model.

Table 4.3: Autoregressive Distributed Lag (ARDL) Model Results

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.001748	0.001788	0.978	0.3309
lag_ExRate	0.1681	0.1043	1.612	0.1107
d_Loss	-0.000011	0.000146	-0.078	0.9381
lag_Loss	-0.000346	0.000146	-2.374	0.0199

Note: Multiple R-squared: 0.1086, Adjusted R-squared: 0.0768, F-statistic: 3.413 (p-value: 0.02113).

The overall model is statistically significant (($F = 3.413$), ($p = 0.021$)), indicating that the explanatory variables jointly contribute to explaining exchange-rate returns.

The contemporaneous internet-loss variable is not statistically significant (($p = 0.938$)), confirming the OLS result that internet disruptions do not exert an immediate effect on exchange-rate movements.

In contrast, the lagged internet-loss variable is statistically significant (($\beta = -0.000346$), ($p = 0.020$)). This finding suggests that changes in internet connectivity influence exchange-rate returns with a delay. The result is consistent with the Granger causality analysis and supports the hypothesis that internet shutdowns affect financial markets through gradual adjustments in expectations and information availability.

The lagged exchange-rate return variable is positive but not statistically significant (($p = 0.111$)), providing only limited evidence of persistence in exchange-rate dynamics. The model explains approximately 10.9% of the variation in exchange-rate returns (($R^2 = 0.109$)), which is reasonable given the complexity of currency markets and the influence of numerous external factors.

To assess whether the Granger causality results depend on the selected lag structure, an additional Granger causality test was estimated using three lags. The results remain statistically significant ($F=7.671$, $p < 0.001$), providing strong evidence that internet connectivity loss contains predictive information regarding future exchange-rate movements. The persistence of significance across alternative lag specifications suggests that the findings are robust and not driven by a particular lag choice. This strengthens the conclusion that internet connectivity disruptions influence exchange-rate dynamics through delayed information-transmission mechanisms.

4.8 Structural Break Analysis

Structural break analysis was conducted to determine whether the January 2026 internet shutdown produced a significant change in the relationship between internet connectivity loss and exchange-rate returns.

The Bayesian Information Criterion (BIC) reached its minimum value in the specification without breakpoints, suggesting that introducing structural breaks does not improve model performance. Although the residual sum of squares decreases as additional breakpoints are added, the BIC indicates that the increase in model complexity is not justified.

These results provide no strong evidence of a distinct structural shift during the sample period. Instead, they suggest that the effects of internet disruption were transmitted through gradual market adjustments rather than through a sudden regime change.

The empirical findings reveal no statistically significant contemporaneous relationship between internet connectivity loss and exchange-rate returns. However, both the Granger causality test and the ARDL model provide evidence that internet connectivity shocks exert a delayed influence on exchange-rate dynamics.

Specifically, past changes in internet loss are found to improve the prediction of future exchange-rate movements, while lagged internet-loss effects remain statistically significant within the ARDL framework. These findings support the theoretical argument developed in Chapter 2, which suggests that internet shutdowns affect financial markets through disruptions to information flows and expectation formation rather than through immediate market reactions.

Table 4.4: Summary of Econometric Analysis

Method	Purpose	Result	Interpretation
ADF	Stationarity	Non-stationary in levels	Variables transformed
OLS	Immediate effect	Not significant (p = 0.261)	No contemporaneous effect
Granger	Predictive effect	Significant (p = 0.018)	Internet loss predicts exchange-rate changes
ARDL	Dynamic effect	Lagged loss significant (p = 0.020)	Delayed impact exists
Structural Break	Regime shift	No evidence	Gradual adjustment

Note: This table summarizes the methodology, purpose, and key findings of the econometric analysis conducted in this study.

Overall, the results indicate that internet connectivity shocks contributed to exchange-rate dynamics in Iran during the January-March 2026 period and provide evidence that digital disruptions may function as economically significant information shocks within financially vulnerable environments.

5. Conclusion

5.1 Summary of Findings

This study examined the relationship between internet connectivity disruptions and exchange-rate dynamics in Iran during the nationwide internet shutdown of

January–March 2026. Drawing on daily network telemetry data obtained from the Internet Outage Detection and Analysis (IODA) platform and free-market USD/Toman exchange-rate data obtained from Bonbast, the research investigated whether internet shutdowns can function as economically significant information shocks within a financially vulnerable environment.

The empirical analysis was conducted using a sequence of time-series econometric techniques, including Augmented Dickey-Fuller (ADF) stationarity tests, Ordinary Least Squares (OLS) regression, Granger causality analysis, and an Autoregressive Distributed Lag (ARDL) model. The results reveal no statistically significant contemporaneous relationship between changes in internet connectivity loss and daily exchange-rate returns. The OLS model suggests that internet-loss shocks alone are insufficient to explain short-run fluctuations in the USD/Toman exchange rate.

However, the dynamic analyses provide a different perspective. The Granger causality test indicates that past changes in internet connectivity contain statistically significant information regarding future exchange-rate movements. Similarly, the ARDL model identifies a statistically significant lagged effect of internet-loss shocks on exchange-rate returns. These findings suggest that the impact of internet shutdowns is not immediate but instead emerges gradually as information constraints, uncertainty, and market expectations influence financial-market behavior.

Overall, the findings support the theoretical framework developed in this study. Consistent with the literature on information asymmetry and exchange-rate microstructure, the results suggest that internet shutdowns may affect currency markets by disrupting the flow of information rather than through direct mechanical effects. Consequently, internet disruptions should be viewed not only as technological or political events but also as potential economic shocks capable of influencing market outcomes.

5.2 Limitations

Several limitations should be considered when interpreting the findings of this study.

First, the analysis is based on a relatively short sample period covering January to March 2026. While this period captures the major internet-disruption episodes of interest, the limited number of observations restricts the statistical power of the econometric models and constrains the ability to examine longer-term effects.

Second, the study focuses exclusively on the Iranian case. Although Iran provides a particularly relevant environment due to its history of internet restrictions and exchange-rate instability, the results may not be directly generalizable to other countries experiencing internet shutdowns.

Third, the period under investigation coincided with significant political and geopolitical developments, including nationwide protests and military conflict. These events may have influenced market expectations, risk perceptions, and exchange-rate movements independently of internet connectivity disruptions. As a result, the observed relationship between internet loss and exchange-rate dynamics may partly reflect broader political and economic conditions that were not explicitly incorporated into the model.

Finally, the limited transparency of economic and financial data in Iran constrains the scope of empirical analysis. Reliable information regarding foreign-exchange interventions, capital flows, market participation, unofficial currency transactions, and other potentially important determinants of exchange-rate behavior is not publicly available. Consequently, it was not possible to construct a more comprehensive model including all relevant explanatory variables. The findings should therefore be interpreted as evidence of a statistically significant association between internet connectivity shocks and exchange-rate dynamics rather than definitive proof of a fully isolated causal relationship.

5.3 Recommendations for Future Research

Future research could extend this analysis in several directions. First, longer time horizons would allow researchers to examine whether the effects identified in this study persist beyond the immediate aftermath of an internet shutdown. Second, incorporating additional macroeconomic and political variables, such as inflation expectations, sanctions-related developments, geopolitical events, and central-bank

interventions, could improve understanding of the mechanisms linking internet disruptions and exchange-rate movements.

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